



Xingyu Wang

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Gender: Male **Date of birth:** 4 Jan 2000 **Nationality:** Chinese

CAREER OBJECTIVE

Career focus

Dedicated Computer Engineer specializing in IoT and AI-driven systems, with strong programming skills in C++, Python. Seeking to leverage my expertise in automation systems and development to contribute to innovative technology projects in a dynamic company.

EDUCATION AND TRAINING

[1 Apr 2022 – Current]

Master of Computer Engineering

Paderborn University <https://www.uni-paderborn.de>

City: Paderborn | **Country:** Germany | **Field(s) of study:** IoT Systems, Embedded Systems, and Machine Learning

[30 Sep 2017 – 1 Oct 2021]

Bachelor of Computer Engineering

Assumption University, Faculty of Engineering <https://www.au.edu>

City: Samutprakarn | **Country:** Thailand | **Field(s) of study:** Computer Engineering | **Final grade:** 3.72/4.00

SKILLS SECTION

Technical Skills

- **Programming & Development:** Python (Advanced), C/C++ (Embedded), OpenCV
- **IoT & Embedded Systems:** Raspberry Pi, Riotee Boards, RIOT OS, nRF52840, BLE Communication Protocol
- **Electrical Engineering & Automaton:** Sensor Integration, Circuit Design, Battery-Free Systems, IT Infrastructure, Edge Computing Architecture

PROJECTS

[15 Jan 2025 – 15 Jan 2026]

Work Experience - Digital Village Twin Etteln (DiDoZ Etteln)

Machine Learning & IoT Project | [Software Innovation Campus Paderborn] - *architect and implement a real-time data pipeline for smart city environments, processing time-series sensor data from different sensors*

As a student research assistant:

- Implemented edge computing architecture to process high-velocity sensor time-series data streams.
- Engineered pattern detection algorithms (DeepAnT - A Deep Learning Approach for Unsupervised Anomaly Detection in Time Series) to dynamically adjust operational modes based on data urgency and identified anomalies.

Technologies: Python, PyTorch, DeepAnT, Streamlit, Real-Time Stream Processing, Edge Computing

[1 Oct 2024 – 1 Oct 2025]

Project Group - Multi-Node Environmental Monitoring Demonstrator

IoT Systems Project | [Paderborn University] - *design and deploy a battery-free IoT network with 4 Riotee sensor nodes, one coordinator, and e-paper display for real-time environmental monitoring*

As a member of three-person team:

- Performed electrical integration including power calculations for energy harvesting systems, capacitor sizing, and cabling for multi-node physical setups
- Developed embedded firmware in C/C++ for ultra-low-power microcontrollers (nRF52840), displaying temperature, humidity, and network status with dynamic updates

Technologies: BLE Protocol, C/C++, Python, Energy Harvesting, Embedded Systems

LANGUAGE SKILLS

Mother tongue(s): Chinese

Other language(s):

English

LISTENING C1 **READING** C1 **WRITING** B1

SPOKEN PRODUCTION B1 **SPOKEN INTERACTION** B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user